1. Title

Feature Selection for Optimizing Glioblastoma Treatment Decisions

2. Group Members

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3. Abstract

Glioblastoma (GBM) is an aggressive brain cancer with limited treatment options. The standard treatment, Temozolomide (TMZ) and Radiation Therapy (RT), does not work equally well for all patients. Some patients respond well and recover, while others experience rapid disease progression despite receiving the same treatment. To further add to the area of personalized treatment plans, this project aims to predict which GBM patients will benefit most from TMZ and/or RT by leveraging single-cell immune and tumor markers. Using feature selection and the appropriate machine learning models, I aim to identify the most important biomarkers that correlate with treatment response, which could optimize treatment strategies for GBM patients by providing personalized treatment selection, avoiding ineffective treatments, and identifying treatment-resistant subtypes.

4. Formal Statement of the Problem

GBM is an aggressive brain cancer with a median survival of approximately 12 to 15 months, despite standard treatments such as TMZ and RT. The two-year survival rate for patients receiving both TMZ and RT is about 26.5%, compared to 10.4% for those receiving RT alone[2]. Currently, there are not any reliable methods to predict individual outcomes as patient responses to these treatments vary significantly, with some patients being able to live upwards of 54 months while others experience quick decline and pass away within a few months[1]. While there are methods such as identifying predictive biomarkers, such as MGMT promoter methylation status, that have shown promise in select demographics in forecasting treatment responses, it's clear that further research in the area will enable personalized GBM therapy, reduced ineffective treatments, and guide alternative therapeutic strategies for patients that are flagged as non-responders to common treatments.

5. Related Work

When looking at the current literature, there has been significant growth in prioritizing individualized treatment plans, including with GBMs. The only source of success in this area when looking specifically at analyzing biomarkers to make predictions would be with MGMT promoter methylation status. MGMT is a DNA repair enzyme which can effectively protect cells against alkylating agents such as TMZ[5]. Therefore, disorders of MGMT promoter methylation are associated with silencing of the MGMT gene, which allows alkylating agents, such as TMZ, to be more effective in patients[5]. Recent randomized trials such as NOA-08, the Nordic trial, and RTOG-0525 have shown MGMT promoter methylation to be a useful predictive biomarker to stratify elderly patients with GBM for radiation therapy versus chemotherapy[6]. As a result, studies have shown that MGMT-methylated patients may have a median overall survival rate of up to 23 months, whereas those without methylation often experience a median survival rate around 12-15 months[6]. Despite these results, the value of MGMT methylation is complicated by evidence that MGMT methylation status can change throughout treatment.

In addition to the MGMT biomarker, there is evidence that certain gene mutations have been linked to positive or negative long-term prognosis. For example, GBM patients who did not have IDH mutations had a median overall survival time of 13 months, compared to 54 months for those that did have IDH mutations[1]. However, this research has not pointed towards a way to individualize patient treatment; rather, it's a helpful tool to determine a patient's prognosis.

Despite all the work in this field, researchers have consistently pointed to a common challenge: GBM tumors are highly heterogeneous at both the genetic and molecular levels[3]. This property makes it difficult at times for a single biopsy to capture the tumor's full profile, therefore making the readings of biomarkers more difficult to interpret.

6. Contributions

My main contributions in this field will be to introduce a technique, feature selection, to identify the most important biomarkers that can successfully predict the correct treatment course for a GBM patient. While it's understood that GBM tumors are highly heterogeneous, the dataset that I am using has hundreds of thousands of cell samples from different periods of time from the same tumor. Therefore, this project should be able to correctly identify the most important biomarkers despite the heterogeneous nature of the tumor. With this work, I expect to show that patients can be screened in a similar fashion as was used to create this dataset, and then receive a more individualized treatment plan that could avoid ineffective treatments and prolong the lifespan of GBM patients.

7. Datasets

I will be using the following dataset from flowrepository: https://flowrepository.org/id/FR-FCM-Z24K. This dataset contains over 2 million cell samples taken from 28 patients. I will also be using the given metadata for the 28 patients.

8. Intended Experiments

From beginning to end, this is how I would achieve the aforementioned goals of this project:

A. Data Preprocessing
    a. Aggregate single-cell markers per patient
    b. Create immune/tumor marker ratio metrics
    c. Normalize and scale data for model inputs
B. Feature Selection
    a. Shapley values - method for fairly distributing the gains and losses among features that has shown success in other research work
C. Train Machine Learning Models to Make Predictions
    a. Logistic Regression
    b. K-Means clustering to stratify patients into treatment-resistant and not treatment-resistant groups
D. Evaluation of Results/Approach

    a. AUC-ROC analysis
    b. Accuracy, precision, recall, F1-score

9. Expected Challenges

There are a few challenges in this project that I expect to be a challenge, which includes:

A. Limited sample size and imbalance data
    a. 28 patients were used to create the dataset, which could lead to treatment response classes (responders versus non-responders) being imbalanced, leading to a biased model
B. GBM is highly heterogeneous meaning a single biomarker signature may not generalize across all GBM tumors for a patient
C. Model may work well on the dataset that I'm using, but it needs to generalize to new patients as well
    a. Reliability of model would then be uncertain
    b. High accuracy on training data but poor performance on new patients

10. Implementation

As aforementioned in the Intended Experiments section, I will provide code at the end that will preprocess the data that then undergoes feature selection to select the most important biomarkers towards categorizing patients into treatment groups. From here, I will train a machine learning model to make the corresponding predictions. The inputs will be the normalized and scaled data from the flowrepository dataset that I linked in the Datasets section. The outputs will be a categorization of patients into the following treatment plans: TMZ + RT, RT, non-responding. I will share this code through the following GitHub repository: https://github.com/RoshanPMahesh/COMP683GBM.git

11. Preliminary Results

I currently don't have any preliminary results. Here is a brief timeline that discusses when I plan to have specific benchmarks of the project completed:

Milestone	Date of Completion
Data Preprocessing	March 26, 2025
Feature Selection	April 9, 2025
Training ML models to make predictions	April 16, 2025
Validate results	April 23, 2025
Create report and presentation	April 28, 2025

The end of the project timeline is variable based on when the final report and presentation are due. Additionally, having a week to validate results gives me the time to go back and fix any errors or tweak the model.

12. References

[1] G. Qin, L. Du, Y. Ma, Y. Yin, and L. Wang, “Gene biomarker prediction in glioma by integrating scRNA-seq data and gene regulatory network,” *BMC Medical Genomics*, vol. 14, no. 1, Dec. 2021, doi: https://doi.org/10.1186/s12920-021-01115-6.

[2] J. P. Thakkar, P. P. Peruzzi, and V. C. Prabhu, “Glioblastoma Multiforme,” *AANS*, Apr. 15, 2024. https://www.aans.org/patients/conditions-treatments/glioblastoma-multiforme/

[3] N. Abdelfattah *et al.*, “Single-cell analysis of human glioma and immune cells identifies S100A4 as an immunotherapy target,” *Nature Communications*, vol. 13, no. 1, Feb. 2022, doi: https://doi.org/10.1038/s41467-022-28372-y.

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[5] S. Shah, A. Nag, Sirpi Vivekanandam Sachithanandam, and B. Lucke-Wold, “Predictive and Prognostic Significance of Molecular Biomarkers in Glioblastoma,” *Biomedicines*, vol. 12, no. 12, pp. 2664–2664, Nov. 2024, doi: https://doi.org/10.3390/biomedicines12122664.

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